

Control Chart Procedure

1 Overview

This procedure explains the concept of control chart and provides entry method to record data into the chart. Control chart shall be used to continually improve the quality of products and the effectiveness of quality management systems.

2 Requirements

Concept of Control Chart:

Manufacturing process fluctuates and is affected by various factors. A control chart is a visual technique used to visualize the control state of a manufacturing process, where data shows temporal changes in certain quality characteristic. It is assumed that a considerable error exists when control chart shows deviations of characteristics from a predetermined control limit value.

Control charts can be classified into two types: control over variable (continuous data) and control over discrete value.

Variable (continuous data) are data of a continuous nature obtained by using calipers, torque wrenches, scales, thermometers, etc. on the object to be analyzed. (e.g., length, diameter, etc. of a part).

Discrete value refer to data that are measurable for the property being analyzed, but have discontinuous properties that can be obtained by using a threaded gauge (no-go gauge) or a master sample, etc. (e.g., pass/fail, number of scratches, etc.)

As a concept of Statistical Process Control, there are two different types of variation that occur in the manufacturing process.

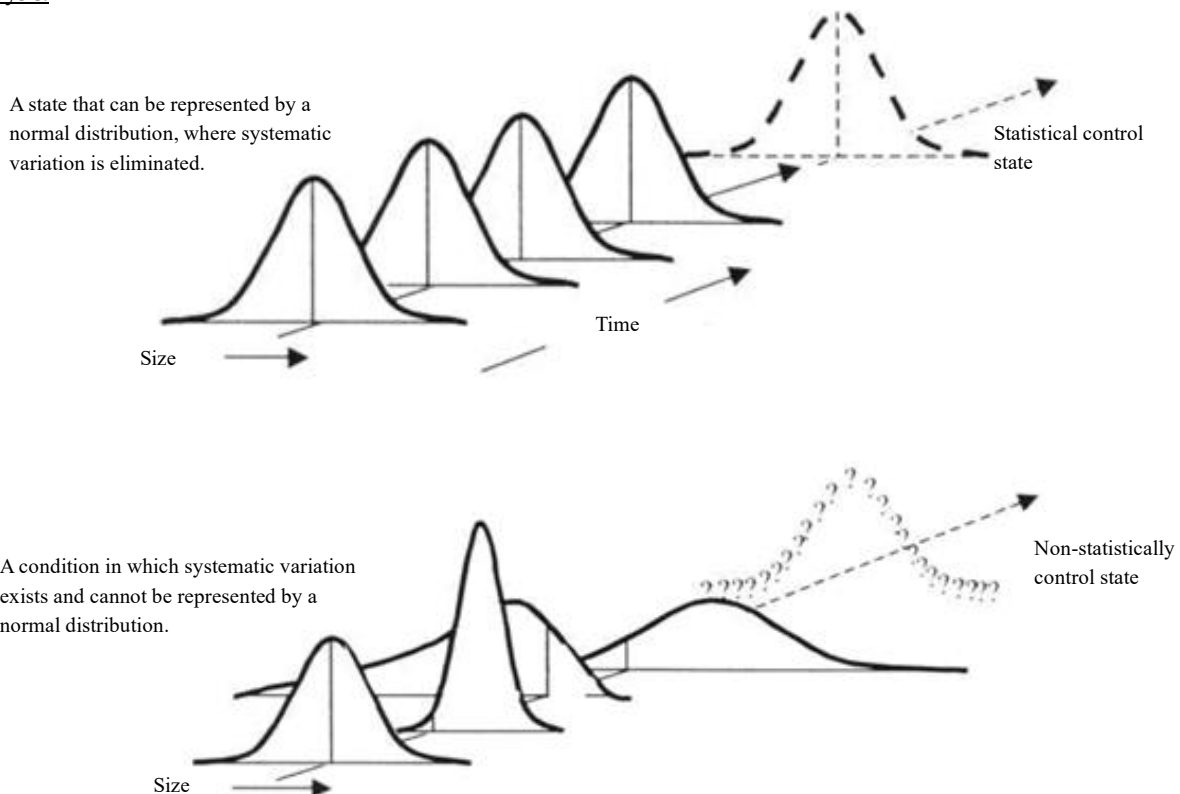
1) Systematic

Refers to a state in which the subsequent distribution of characteristics, product acceptance rate, etc. can be predicted. (e.g., variations that are systematically repeatable, such as cutting tool wear, etc.) Systematic variations are occasional or transient variations that should be identified and eliminated.

2) Random

A contingent variation that cannot be predicted and refers to a condition that can be represented by a normal distribution (e.g., dimensions of machined parts). (e.g., dimensions of machined parts, etc.) Since random variation cannot be eliminated, statistical control is necessary.

The suppliers should aim for a statistical control state where systematic variation is eliminated and only random variation exists. In a statistical state of control, its manufacturing process is stable over time and has a normal distribution, allowing for precautionary analysis.



2.1 Control Chart

Control chart is a chart (line chart with control limits identified) used to investigate (process analysis) or monitor (process control) stability of a manufacturing process that determines with quality characteristics.

Control chart shall be used not to discover nonconformities but to prevent problems from occurring. (This is on the basis that specification limits are outside the control limits.)

2.2 Types and Choices of Control Charts

There are various types of control charts, while all are fundamentally similar. Appropriate chart shall be selected for each purpose based on the subject matter of control.

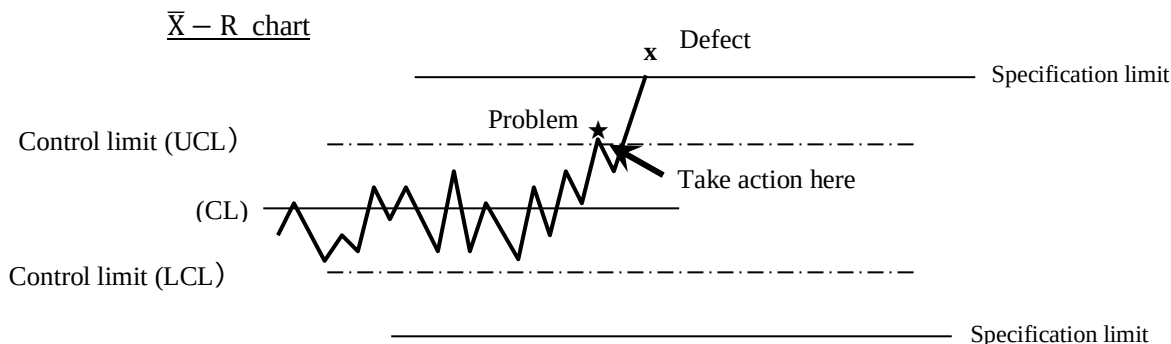
Selection of control chart depends on the type of data used, i.e. continuous or discrete data.

Type of data		Type and outline of control charts	
Variable (continuous data)	Length Weight Time	$\bar{X} - R$ chart (X bar R chart)	A representative control chart for continuous data process control and it displays the maximum and minimum difference values (R) between the mean and the data. Usually analysis is done using a set of control charts as pair of which one is for control of data distribution and the other is for control of deviations. This is the most widely used control chart.
	Strength Constituent Yield	$\tilde{X} - R$ chart (X median R chart)	An alternative to the X bar and R chart and used to control processes from which measurement data will be obtained. A control chart using X (median: the central value in a distribution) rather than X bar and displays middle value ($\tilde{X}$) and range (R). The median of a set of data arranged in order of increasing or decreasing magnitude is expressed by $\tilde{X}$.
	Purity Filled content, etc.	X chart	The X control chart is a control chart which plots a set of data points for each calculated value. It allows immediate plotting of data where grouping of data is not practical or where data is obtained one by one over a certain period of time due to batch production. This X chart has the characteristics that it does not cause a delay in time from the point of obtaining data to judgment and response to process conditions.
Discrete value	Percent defective	p chart	The p chart is used to calculate the mean percent defective in a certain period of time. The percent defective awakens operators or management personnel to the changes in the rate.
	Number of defects	np chart	The np chart is used when the number of defects in the sample is expressed by np and when n, which is used to express the number of products generally contained in the sample, is a fixed number. The np-chart is a control chart used to study mainly the problem rate and number of defects as the p chart.
	Number of problems	c chart	A control chart which is used where sample size of data (area, length, etc.) is fixed to control processes by using values from the discrete data such as the number of problems, problem points, accidents, and repair, etc.
	Number of problems per unit	u chart	A control chart which is used to control processes by using data from the discrete data as with the c chart. The U chart is used where the sample size of data is not fixed.

2.3 Representative Control Charts

2.3.1 $\bar{X} - R$ Chart

$\bar{X}$ (Xbar) chart is mainly used to monitor changes in mean values (between-group variation: dispersion between groups), and R-chart is to monitor changes in the dispersion of values (within-group variation: dispersion within a group). Xbar-chart and R-chart are frequently used in combination.

2.3.2 Preparation of $\bar{X} - R$ chart

2.3.2.1 Collection of Data

Collect as much data as possible about the characteristics which provide important information of the process. Ensure that collected data is comparatively new, is applicable to the process in the future, and have a clear record.

2.3.2.2 Grouping

Group data based on data history, and sort by lot sequence and sampling measurement sequence etc.. The number of data in one group (the sample size) is generally expressed by n, and the number of groups, by k.

2.3.2.3 Data sheet

Create a datasheet with consideration of section 3.3.2.1 and 3.3.2.2. and enter data. Examples of datasheets are provided in section 7. No specific style of form is designated.

2.3.2.4 Calculation of Mean Value ($\bar{X}$)

Calculate the mean, round to one more decimal place than the measured value.

Find the mean value ($\bar{X}$) per group.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \cdots X_n}{n}$$

2.3.2.5 Calculation of Range (R)

Find the range (R) per group.

R is used to indicate dispersion within a group and is a difference between the maximum and the minimum values within a group.

$$R = X_{\max} - X_{\min}$$

2.3.2.6 Calculation of Grand Mean ($\bar{\bar{X}}$)

Add $\bar{X}$ of all groups and divide by the number of groups k.

$$\bar{\bar{X}} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \cdots \bar{X}_n}{k}$$

Calculate the mean to two more decimal places than the measured value.

(Do not round off since it will be used in later calculations.)

2.3.2.7 Calculation of Range Mean Value ($\bar{R}$)

Add all the R values of each group and divide by k.

$$\bar{R} = \frac{R_1 + R_2 + R_3 + \dots R_k}{k}$$

Calculate the mean to one more decimal place than the measured value.

(Do not round off since it will be used in later calculations.)

2.3.2.8 Calculation of Control Limits

$\bar{X}$ control chart

Center line: $CL = \bar{\bar{X}}$

Upper control limit line: $UCL = \bar{\bar{X}} + A_2 \bar{R}$

Lower control limit line: $LCL = \bar{\bar{X}} - A_2 \bar{R}$

R chart

Center line: $CL = \bar{\bar{R}}$

Upper control limit line: $UCL = D_4 \bar{R}$

Lower control limit line: $LCL = \bar{\bar{x}} - 3\sigma_{\bar{x}}$ ignored when n is equal to or smaller than 6.

Control limit line coefficient		
n	A_2	D_4
2	1.880	3.267
3	1.023	2.575
4	0.729	2.282
5	0.577	2.115

2.3.2.9 Entry of Control Lines

Set scale on the control chart so that the spacing between the upper and lower control lines will be about 20 to 30 mm and enter CL, UCL and LCL on the chart. Draw center line (CL) by solid line and draw limit lines by broken lines in the case of data analysis or dashed-dotted lines in the case of process control.

Centerline	:	solid line	(—————)
Limit line	{	analysis : broken line	(- - - - -)
		process control : dashed-dotted line	(-. . -. .)

In case the control chart for analysis is extended for application to process control, the line is changed from broken line to dashed-dotted line. Also enter CL, UCL, LCL and each value near the lines.

2.3.2.10 Plotting

For plotting normally, “●” is used for X and “x” is used for R. Plots outside of the control limits (including those on the limits) should be so marked as to facilitate identification. (e.g. “◎” or in red, etc.).

2.3.2.11 Points to Remember

In general, set control limits based on the dispersion of data when preparing a control chart.

In some cases control limits are set at the upper specification limit and the lower specification limit with an optimistic view of no nonconformance being expected. However if control limits are set at the specification limits, the chart will not allow detection of possible abnormality and it is a misuse of the chart.

Count the dispersion of data in setting control limits (action lines) on a chart when using specification limit. Use the chart to prevent problems from occurring by taking precautions when dispersion is observed in the red zone.

2.3.3 p chart

p chart does not require a fixed sample size. P chart is used in control of the process in which percent defective can be expressed in percentage. For instance, when finished- or semi-finished goods of 100 sheets, 100 units or n in general are inspected for quality, 5 (r or np in general) out of 100 of products are found to be non-conforming, therefore the percent defective P is $5/100$, .05 or 5%. The percent non defective may also be used.

2.3.4 Preparation of p control chart

2.3.4.1 Data collection

Collect data of inspection results with number of units inspected (n) and number of units rejected (np). Minimum of 20 sets of data shall be collected.

Data shall be collected and recorded with respect to each lot, machine, date, time, etc.

2.3.4.2 Grouping

Form a group with the collected data. The size of a group n should not be too large or too small since groups with many p=0 points are not appropriate.

2.3.4.3 Determination of Percent defective (p)

Fill in a data sheet with the group number, n, np, etc., and calculate a percent defective for each group (p).

$$p = \frac{\text{no. of rejects}}{\text{size of a group}} = \frac{np}{n}$$

Multiply by 100 for percentage.

2.3.4.4 Determination of Average Percent defective

$$\bar{p} = \frac{\text{total of rejects}}{\text{total of units inspected}} = \frac{\sum np}{\sum n}$$

(note that this is not the mean value of the percent defective p of each group.)

2.3.4.5 Calculation and Entry of Control Limits

Center line: CL=p

$$\text{Upper control line: } UCL = \bar{p} + \frac{3\sqrt{\bar{p}(1-\bar{p})}}{n}$$

$$\text{Lower control line: } LCL = \bar{p} - \frac{3\sqrt{\bar{p}(1-\bar{p})}}{n}$$

2.3.4.6 Control Chart Preparation

2.3.4.7 Points to Remember

If the sample size n varies among groups, center line will remain the same, but control limits shall be calculated for each group and a limit value be applied for each point.

Thus, limit lines on the control chart may be uneven, and control range becomes smaller as n becomes larger. If n is larger than 500, use a stratification approach since the control range becomes exceptionally small.

2.4 Reading of Control Charts

In order to understand the condition of processes i.e., acceptable, marginal or Important, it is necessary to read trends in data points appeared on control chart. The purpose is to take actions proactively when abnormality is observed in control chart.

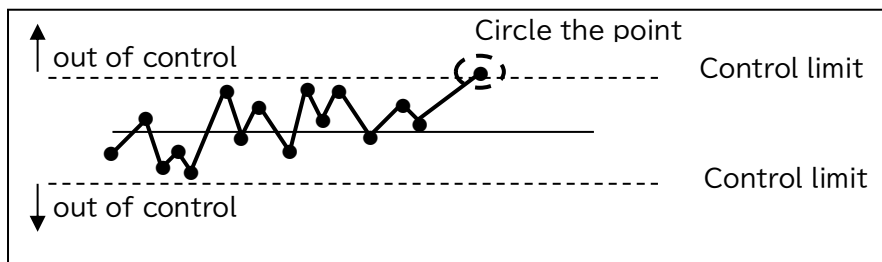
2.4.1 The criteria for judging process conditions, whether or not a process is in a controlled state (or stable state), are as follows. It is advisable that evaluations be made on the basis of at least 25 plotted points.

2.4.1.1 No plotted points outside control limits: no out-of-control points

However, a process is considered as being in a controlled state if the following cases apply.

- 1) When 25 consecutive points are within the limits.
- 2) One point or less out of 35 consecutive points lies outside the limits.
- 3) (And, this one point exhibits no particular abnormality.)
- 4) Two points or less out of 100 consecutive points lie outside the limits.
- 5) (And, these two points exhibit no particular abnormality.)
- 6) No bias in sequence, trends, cycles, etc. in the order of plotted points.

2.4.2 Data Point Exceeds Control limits

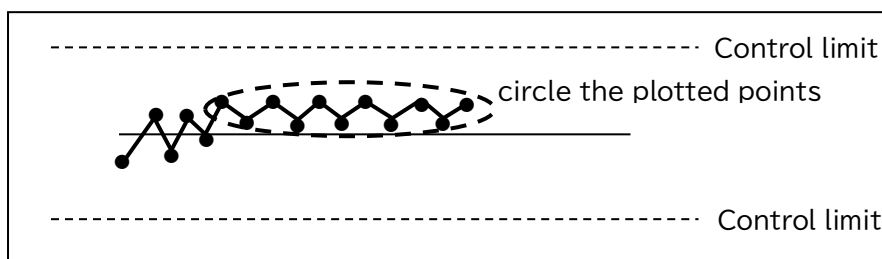


The points outside control limit lines are referred to as being “control-out” or “out of control”. Reasons for points being control-out are as follows.

- 1) Due to a random cause resulting in the point lies outside the limit lines, or
- 2) An abnormality has occurred in the process and resulted in the point being outside the control limit lines.

Phenomenon described in (1) above may occur with a probability of three times in 1000 plots. If control-out occurs, take action on the premise that 2) above is most likely to have occurred and probe the cause of abnormality.

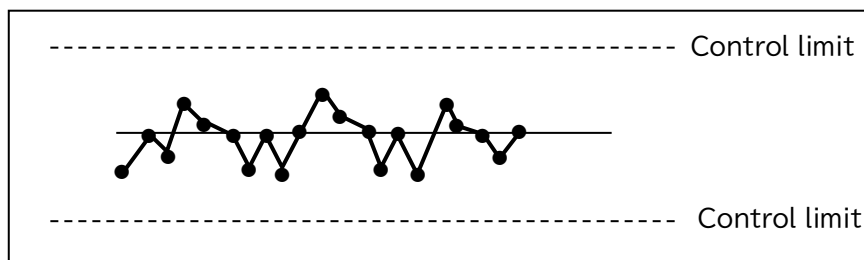
2.4.3 Run of Data Points



A series of points appear consecutively on one side of the centerline (median line, to be precise, may be substituted by the centerline) is called a run. The length of a run is the number of consecutive points to one side of the centerline.

A run with seven or more points is judged as indicating control-out state.

2.4.4 Data Points Show a Cycle

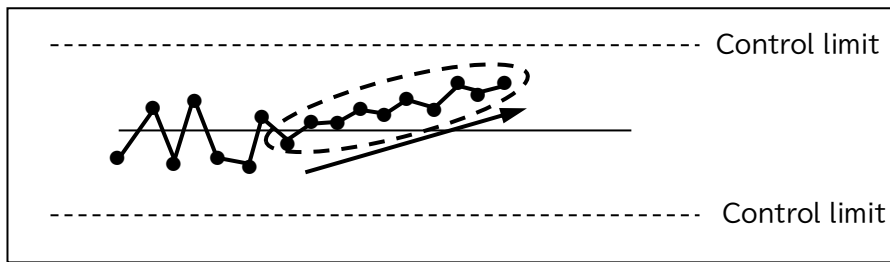


When data points show a cyclic pattern of always declines on Mondays, there must be a reason.

Investigation into the cause of cycles peculiar to the process is very useful in terms of process analysis. Although these cycles are not easy to identify, it is ideal to analyze them with experiences and engineering knowledge over a long period of time and develop control methods well matched to the rhythm (cycle).

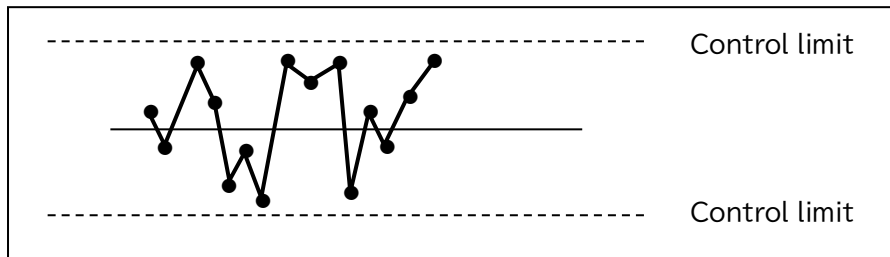
It is necessary to discern the cause of the cycle and carefully evaluate the process condition to determine whether or not a process is in a controlled state.

2.4.5 Shows Trends (Upward or Downward)



If there is an upward or downward run of consecutive seven points or more, it is also judged as out of control. In case of a steep rising or falling, take immediate action to investigate the cause when any obvious trends in the plot of data is observed.

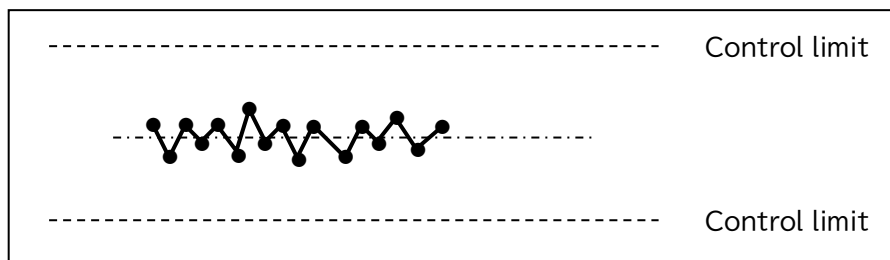
2.4.6 Two Spikes on Chart



It is also considered abnormal if there are two spikes appear in the plot of raw data. There may be a mix up of data of two or more.

(e.g.) two spikes have different patterns, etc.

2.4.7 All Points Gathered Around the Center Line



If the process is in a controlled state, the plotted points must be distributed randomly.

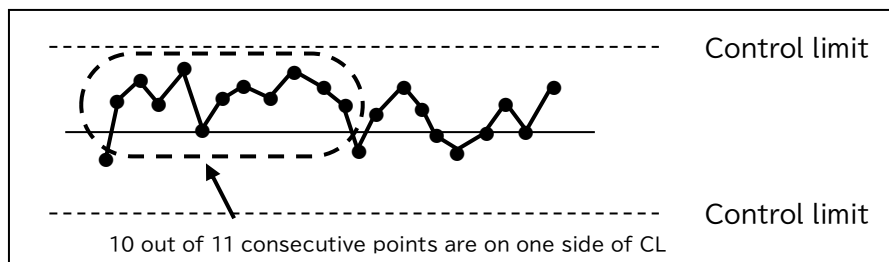
When considering distribution confirmed by control limit lines at $\pm 3\sigma$, the slope down gradually becomes lower from the center to the edges so that points are less likely to occur about the center line when they get close to the control limits. Compared to the vicinity of the centerline, points appear less readily as the control limits are approached. In other words, points should not occur concentrated about the center line.

It is often the case that there is either a problem with the grouping, or groups contain abnormal data.

It is also judged abnormal if 15 or more consecutive points fall between $\pm\sigma$.

(e.g.) measurement error.

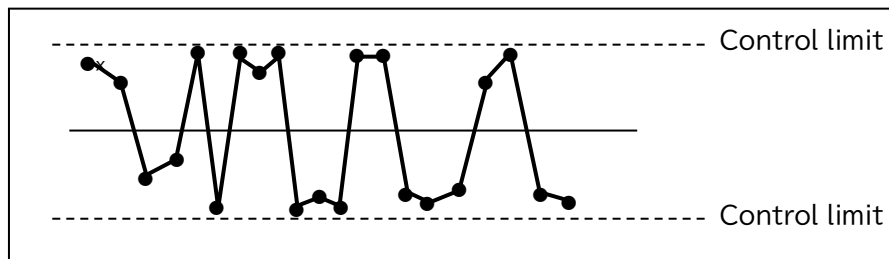
2.4.8 Consecutive Points on One Side of the Center Line



If one of the following conditions appear on one side of the centerline (median line to be precise), the process is judged as out of control controlled state.

- 1) 10 or more points out of 11 consecutive points
- 2) 12 or more point out of 14 consecutive points
- 3) 14 or more point out of 17 consecutive points, or
- 4) 16 or more point out of 20 consecutive points.

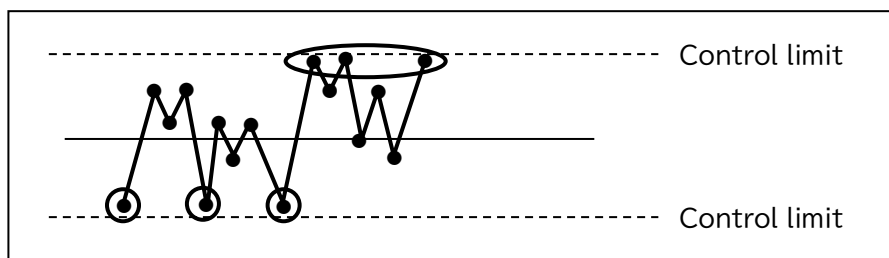
2.4.9 Few Points along the Center Line



Under normal circumstances, a large number of points should be near the centerline if the sample follows a single normal distribution. Accordingly, the chart above shows an abnormality.

This is often found when plotting data on the chart with sample groups taken from each of two machines with different process average. It is necessary to stratify the sample groups and draw control chart for each machine.

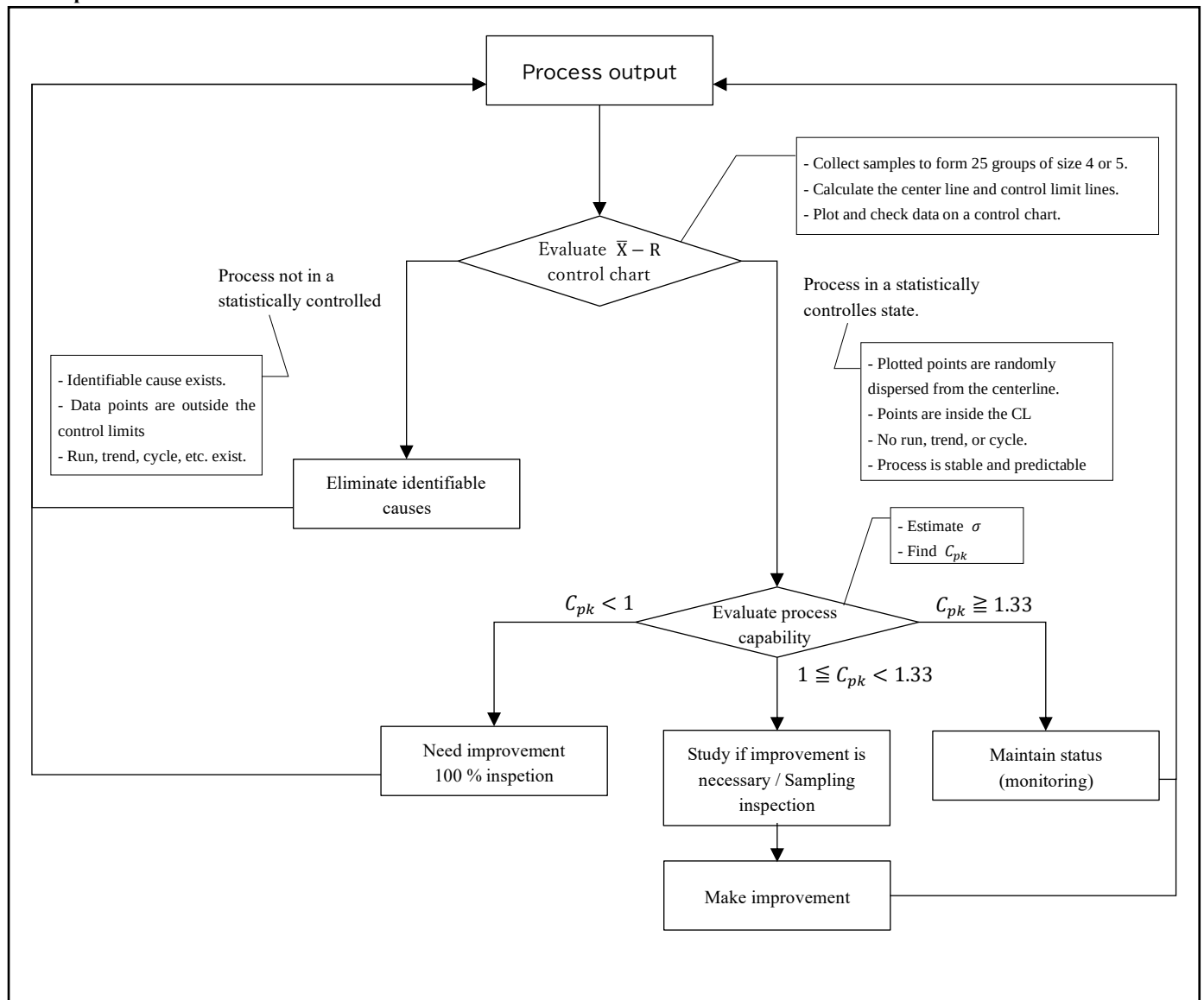
2.4.10 Points Close to the Control Limit Lines (Upper and Under)



Evaluate as control-out in the following cases when points are near the control limit lines, beyond the line two third of the distance from the center line to the control limit lines.

- 1) 2 out of 3 consecutive points
- 2) 3 out of 7 consecutive points
- 3) 4 out of 10 consecutive points

The above conditions shall be judged abnormal since probability of the process being beyond $\pm 2\sigma$ range from the centerline is approximately 5 percent.

3 Operation flowchart

Part Name		Part Name		[P Chart] Date: YY/MM/DD																			
-																							
Description		(%)																					
Item		Date No. of units																				Total	
Number processed																							
Number of defective units																							
Defect rate (%)																							
Notes																							
Be sure to provide the following information. 1. Causes and measures taken if points had been found lying outside the control limits. 2. Specify changes if there is a change to the contents of a control chart. 3. Any changes to operation standards, or to other major items which require changes																							

Original to be retained until YY/MM/DD